Example : $F = \{A \to CD\}$ then $\{A \to C, A \to D\}$

- FD set must be non-redundant FD. (eliminating unnecessary FD's)

Example : $F = \{A \to B, B \to C, A \to C\}$ then

$F = \{A \to B, B \to C\}$ because $A \to C$ is redundant.

> Note:
>
> Minimal Cover of FD Set (F) may not unique, but all minimal cover's logically equal.
>
> $\therefore \quad \boxed{Fm_1 \equiv Fm_2 = F}$

1.8 Normalization

Normalization used to eliminate/reduce redundency in DB relations.

The normalization is especially meant to eliminate the following anomalies:

- Insertion anomaly
- Deletion anomaly
- Update anomaly
- Join anomaly

1.8.1 Normalization of DB table

Decompose relation into two or more sub-relations in-order to reduce or eliminate redundancy and DB anomalise.

1.8.2 Properties of Decomposition

- **Loss-less Join decomposition:** Relational Schema R decomposed into R1, R2, R3, ... Rn sub-relations.
- In general $[R1 \bowtie R2 \bowtie R3 \ldots\ldots\ldots Rn] \supseteq R$

1. If $[R_1 \bowtie R_2 \bowtie R_3 \ldots\ldots \bowtie R_n] \equiv R$ then loss-less join decomposition.
2. If $[R_1 \bowtie R_2 \bowtie R_3 \ldots\ldots\ldots \bowtie R_n] \supset R$ then lossy join decomposition.

Example :

Let R be the relational schema decomposed into R_1 and R_2.

Given decomposition is loss-less join if–

1. $R_1 \cup R_2 = R$ (all attributes covers)
2. $R_1 \cap R_2 \neq \phi$
3. $\underbrace{R_1 \cap R_2 \to R_1}_{R_1 \cap R_2 \text{ is SK of } R_1}$ or $\underbrace{R_1 \cap R_2 \to R_2}_{R_1 \cap R_2 \text{ is SK of } R_2}$

1.8.3 Dependency Preserving Decompostion

Relational Schema R with FD set F decomposed into $R_1, R_2, \ldots\ldots\ldots R_n$ sub relations

Assume $F_1, F_2, \ldots\ldots\ldots F_n$ FD sets of sub relations respectively.

In general

$[F_1 \cup F_2 \cup \ldots\ldots\ F_n] \subseteq F$

1. If $[F_1 \cup F_2 \cup \ldots\ldots\ F_n] \equiv F$ then dependency preserving decomposition.
2. If $[F_1 \cup F_2 \cup \ldots\ldots\ F_n] \subset F$ then not dependency preserving decompostion.

1.9 Normal Forms

Used to find degree of redundancy

Redundancy in relation because of

Non-Trivial FD
$x \rightarrow y$
(Single value dependancy)

Non-Trivial MVD
$x \rightarrow\rightarrow y$
(Multivalued dependancy)

- BCNF relations have 0% redundancy over FD's where as 4NF relations have 0% redundancy over FD and MVD.
- To Eliminate redundancy over Non-Trivial FD, relation should decompose into BCNF but BCNF may not avoid the redundancy due to MVD.
- To eliminate redundancy over non-trivial MVD, relation should decompose into 4NF (4^{th} Normal Form).

1.9.1 First Normal Form

- Default NF of RDBMS relations.
- Relation (DB Table) R is in 1 NF if and only if no multivalued attributes in R. [Every attribute of R must be atomic/single valued].

Note:
Degree of redundancy is very high in 1NF relation.

Important Point 1

- $x \rightarrow y$ FD forms redundancy in relational schema R if and only if –
  (i) Non-Trivial FD
  and
  (ii) X is not super key.
- $x \rightarrow y$ FD not forms any redundancy in relation R if and only if–
  (i) Trivial FD [$x \supseteq y$] or Semi-trivial
  or
  (ii) x : super key

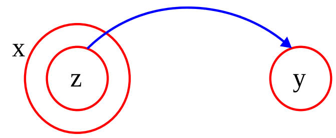
1.9.2 Second Normal Form (2NF)

Relational Schema R is in 2 NF iff
- R should be 1NF
- R should not contain any partial dependency

Partial Dependency

Let R be the relational schema and x, y, z are non-empty set of attributes.

X : Any candidate key of R.
Y : Non-prime attribute
Z : Proper subset of candidate key



∴ z → y is said to be partial dependency iff–

- z is proper subset of candidate key
- y should be non-prime attribute.

Important Point 2

$$\underbrace{[\text{Proper subset of Candidate key}]}_{\text{Not super key}} \rightarrow [\text{Non-Prime attribute}]$$

The above FD forms redundancy in R.

1.9.3 Third Normal Form (3NF)

Relational schema R is in 3NF iff every non-trivial FD x → y in R with–

(i) x must be super key (SK)
or
(ii) y must be prime attribute.

∴ {SK → Prime/Non-Prime, (Not SK) → Prime attribute}

Important Point 3

3NF allow FD set:
[Proper subset of candidate key] → [Proper subset of other candidate key]
which forms redundancy in R.

1.9.4 Boyce codd Normal Form (BCNF)

Relational schema R is in BCNF iff every non-trivial FD "x → y" with x must be super key.
∴ Prime/ Non-prime attributes must be determine by super key.

Important Point 4

If R is in 3NF but not BCNF then [Proper subset of candidate key] → [Proper subset of other candidate key] must exists in R.

Important Point 5

If relational shcema are with only simple candidate key then R always in:

I. 2NF
II. May or may not 3NF/BCNF

Reason : [Proper subset of candidate key] → [Non-prime attribute]
From the above statement, we can conclude that "partial dependency" not possible if all CK's are simple candidate key.

Important Point 6

If relational schema R with only prime attribute (No non-prime attribute in R) then R always in:

I. 3NF
II. May or may not BCNF

Important Point 7

If relational schema R with no non-trivial FD's then R always in BCNF.

1.9.5 Database design Table

DB design goal	1NF	2NF	3NF	BCNF
1. Loss-less jon decomposition	Yes	Yes	Yes	Yes
2. Dependency preserving decomposition	Yes	Yes	Yes	May not
3. O% redundency	No	No	No	Yes [Over FD]

- Every relation possible to decompose into 1NF, 2NF, 3NF, BCNF with loss-less join decomposition.
- Every relation possible to decompose into 1NF, 2NF, 3NF with Dependency preserving decomposition.
- Not every relation can decompose into BCNF and 4NF with dependency preserving decompostion.
- Most accurate normal form to design simple database is 3NF because every relation is possible to decompose into 3NF with loss-less join and dependency preserving.

❑❑❑